

Tab 1

Keep On Track: Element C

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Design Requirements Numbered in Priority

1. Cause and Effect: HIGH Priority

This design shall display cause and effect functions that aid in expanding life skills for children with special needs. The rationale behind our number one priority is due to the benefits that it will provide. The Lifeskills class at Whitemarsh Elementary wants a piece of assistive technology that will unravel new, complex ideas and information in a more simplistic way for an overall understanding (Australian Disability Clearinghouse on Education and Training). Simple, everyday cause and effects are light switches, car ignitions, or something less tangible like not wearing hats and gloves in the winter so the skin turns purple. The processing of these concepts are a challenge but repetition is crucial for comprehension. Five or less cause and effect functions will be the minimum, so anything less will count towards an unsuccessful requirement. We can test the cause and effect function by simple survey and assessment. What lessons do we want children to take away? Can they identify the cause and effect? Is the cause and effect straightforward? How does the chosen cause and effect function relate to life skills and social cues? Assessing the why, what, how will lead to impressive results that convey true developmental progress.

2. Accessible: HIGH Priority

This design shall be accessible to any child with special needs to enrich the overall lesson of developing critical skills for real-world application. The students in the Lifeskills class have a wide variety of needs, wants, and behavior so accessibility is essential. The goal of accessibility also spreads awareness for less fortunate children that could grow mentally if provided with assistive technology. UNICEF has estimated that only 1 out of 10 children who come from financially restricted backgrounds have

the means to obtain assistive technology (UNICEF; PMC9915405). This reasoning sets accessibility as our second highest priority: to show others that every child has the right to learn from this type of technology. We know our product is successful if kids with autism, down syndrome, ADHD, color or hearing impairments, and many other disabilities can all use the device and learn from the intended developmental mechanisms.

3. Safety: HIGH Priority

This device shall prevent any bodily harm to the children interacting with the device. From the words of Mrs. Brennan, the children in the Life Skills class have a tendency to throw objects around the classroom, this stresses our reason why this is our third highest priority. Emotional regulation is difficult for certain children and possible events such as throwing toys in frustration are one of factors that feed into safety concerns. Sanitation is another must, the surface of the device needs to be disinfected, this will decrease the spread of germs in the classroom. The materials have to be non-toxic and any pieces/parts can't be small in size to be considered a choking hazard. On the topic of hazards, Medical experts at KidsHealth inform that baby and toddler toys should be "1¼ inches in diameter and 2¼ inches in length". A simple test for choking hazards is to compare the toy's size to the tube of a toilet paper roll, anything that fits in the hole is non-negotiable and should not be played with by a baby or toddler. When the final rendition is complete and presented to the Lifeskills class, any later report of an injury is a failure. It is the group's duty to safety check the materials—drop, bend, twist, chew. Any cracks or sorts that result from our testing is a no-go and will no longer be a consideration. Furthermore, if there are any wiring components, they must be hidden or blocked off. With all things considered, the brainstorming and manufacturing process must be carried out with care to ensure that the children can play safely.

4. Sensory: HIGH Priority

The bells and whistles of any toy are the aspects that make the toy very meaningful to children. We want our device to encapsulate the mystics behind the fun of toys, while still serving as an assistive piece of technology for early childhood development. This device shall cater to the five senses: sound, taste, touch, sight, and smell. For this requirement to be successful, our device needs to include components that establish at least three out of the five senses. The website *Autism Speaks* informs that “Sensory challenges can affect the student’s ability to take in information, respond to requests, participate in social situations, write, participate in sports, and maintain a calm and ready to work state” (Autism Speaks, 2018). Any sensory space should be a sanctuary for children wanting to escape from triggering noises or other disturbances. As to make the device more personal for the children at Whitemarsh, we can ask questions like favorite colors? certain sensitivities? Calming noises? The list can go on and on. Along with achieving three out of the five senses, we know we are successful if the children display a calmer state after playing with our assistive technology. All in all, children deserve safe spaces to ease distress and that is why sensory regulation is a top priority.

5. Engaging: HIGH Priority

This design shall engage students with designated sensory features, which will keep them occupied for the duration of the intended play time. Engagement is a must-have goal to keep in mind when designing a device for elementary-aged children. Minimal to zero distractions is desirable but children with physical and mental disabilities are a bit sporadic so as a testing method, we want to time attention spans of the children when they play with other assistive technology devices. A good, manageable window for engagement time is between 10-15 minutes. This is plenty of time for the kids to destress and play around with the toy, it will too, help the teachers escalate the situation (if there is any). Engagement ties in with independency also, this societal skill builds from solo activity play, which in turn will boost performance, development, and quality of life (PMC9915405). With all reasons and metrics stated, this is why engagement is one of our five top priorities.

6. Durable: LOW Priority

This design shall be durable to maximize longevity for the sake of the teaching staff. The goal behind this requirement is to guarantee structure durability so that nothing has to be repaired in the short term. The majority of teachers have limited knowledge and skill of things like Arduino, electrical wire, and motors so they would be incapable of fixing broken bits. Testing out the life-span of certain coding circuits, wiring, and motors beforehand would ensure that these components are durable for continual usage. Another useful tactic is reading product reviews of any material or object we buy online, ratings are a personal insight on the pros and cons of the item. The goal is to ensure the device contains the structural integrity for more than 5+ years—this requirement is more for the physical structure. This device would be deemed a failure if the structure breaks within one use. To help out Mrs. Brennan and Dr. Sabatino, our group can always create a guide that comes along with the device to demonstrate step by step any complex components that might need maintenance. This solution will ease their worry of dealing with devices that are a bit out of their “fixing range”. Although we can not completely verify the life-span of the device, the priority is low as we rather prioritize safety.

7. Inexpensive: LOW Priority

This design shall be specific to our group's budget and not real-world application to our future assistive technology. Inexpensive is a low priority on the account of the materials (wood, machinery, CADD softwares) being accessible for our usage. This application of the inexpensive requirement is a group goal— to not surpass the budget quota and to indirectly combat the inaccessibility factor of assistive technology. Many parents, hospitals, and under-funded school districts struggle in a collective battle against a saturated market of expensive toys/devices (Langtree). The logistical goal is not to surpass our budget, so any dollar more spent is contradictory to our low-cost endeavors. Realistically speaking, this product is only being designed and built for Whitemarsh so there are no real stakes if we exceed the budget. But in the context of

providing development-specific technology to a group of children facing disabilities, we want our device to be resemblant of a cost effective device. This goal ensures that, in real world application, our device would be available to any buyer.

8. Aesthetics: LOW Priority

This design shall be visually appealing to the eye by use of color, texture, material, lighting, and other components. Aesthetics is rated a low priority because of possible timing issues, there is only a semester to complete the project so by the end of the project, there might be more urgent matters. But aesthetics is still a requirement, the device can not be bare: no finish, no paint, no design. All these artistic details help with pattern recognition, establishing further the idea of cause and effect repetition. In correlation with our primary stakeholders, they brought up color contrast, which they said will help children with vision impairments, this factor justifies why aesthetics are a requirement for the assistive technology. To be successful, there simply needs to be a purpose behind the layout, the color choice, design, size, shape, material—all aimed with the intention for life skill enhancement.

Primary Stakeholder Validation

Our first visit to Whitemarsh Elementary School was a successful trip to further explore ideas that will aid in cause and effect. Mrs. Brennan and Dr. Sabatino laid out the groundwork for the design: easy for the teachers to work, leisure engagement, less iPad focused, and possible lights feature. They both clarified that it was *our* project and they were content with any ideas for a device that helped with developing life skills. Our biggest takeaway was their focus on simplicity, not in terms of the design instead, something that requires installment and/or calibration. Dr. Sabatino told us how she wishes the device would be more of an independent activity rather than teacher led. Our chosen requirements cater to an assistive technology that builds upon cause and effect toys but stray away from using iPads. It is both our goal and the teachers to get the kids off screens and play with more tangible toys. Many of the kids in the Lifeskills class have been hospitalized for long periods of time so it created a dependency on screens so

this new assistive technology will break this barrier. During our visit, the team mentioned manual facets that advance motor skills in children with disabilities. We gave sample solutions such as on/off switches, a hand crank, colored light remotes, anything that requires hand movement essentially. Another topic of discussion was disinfection, Brennan and Sabatino stressed the importance of sanitation which is a sub-requirement mentioned in our safety priority. Overall, the group and the teachers are on the same boat: to make a piece of assistive technology that promotes cause and effect development with tangible components that include sensory, manage engagement levels, bring a bit of color to the sensory space, and is accessible for all the unique needs of the Lifeskill students.

Complement of Test Methods

	Accessible	Aesthetics	Durable	Engaging	Inexpensive	Safe	Sensory
Test Method/ Goal reaching	-can be accessible for wheelchair users -spatial room to be a duo or individual activity -Motor-skill developmental mechanics are fully operative	-test color gradients for best color to aid in color-eye coordination -double check for any pumps, ridges, grooves, or any other hindrances in the toy	-Read customer reviews of any components that will be bought on online -Test out 3D filaments (price, toxins, flexibility)	-Time the attention span of Life Skills kids interacting with other toys	-Use scraps of material for prototypes -Compare prices of other assistive technology	-Twist, bend, chew, throw etc any part -Test to see if disinfecting surface is possible	-Ask the children favorite sounds, smells, colors -Test out light settings as effects like strobe are too harsh

dont grade

<u>Requirement</u>	<u>Priority</u>	<u>Rationale</u>
Accessible	TOP	
Sensory	TOP	
Safe	MEDIUM	
Engaging	MEDIUM	
Durable	LOW	
Inexpensive	LOW	Inexpensive is a low priority on the account of access that we have to materials such as wood, machinery, CADD, etc. In the context of accessibility of assistive technology, our group wants the manufacturing side to be cost effective to ensure that, in real world application, the marble wall would be available to any buyer. this ensures the hypothetical idea of The reason why it's low is due to the fact that we already have access to wood from the school shop and utilizing an at-home 3D printer.
Aesthetics	LOW	

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. design requirements are related to prior research.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 No
Notes • Your design requirements should be tethered to all that you've written in Element A. This subelement measures how many of your design requirements are related to the research you did and the information you found in Element A.						
C2. Requirements list. Design requirements _____.	score = 5 are listed, prioritized, clear, and detailed	score = 4 are listed, prioritized, generally clear, and detailed	score = 3 are listed, prioritized, generally clear, and somewhat detailed	score = 2 are listed and prioritized but some or all may be incomplete and/or lack specificity	score = 1 are listed, formatted, and prioritization is attempted, but these may be partial and/or overly general	score = 0 are not present or are too vague
Notes • You must give a numbered prioritized list (and stated that it is prioritized) to get higher than a 0 AND state whether each requirement is high or low priority.						
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5 Five	score = 4 Four	score = 3 Three	score = 2 Two	score = 1 One	score = 0 None
C4. Objective & measurable. Design requirements _____.	score = 5 are consistently objective and measurable	score = 4 are nearly always objective and measurable	score = 3 are generally objective and measurable	score = 2 may be only sometimes objective and/or measurable	score = 1 are insufficiently measurable	score = 0 are not measurable
Notes • Each design requirement needs to be measurable so one can objectively assess whether it meets the objective or not.						
C5. Leading to solution. Design requirements _____.	score = 5 would be highly likely to lead to a tangible and viable problem solution	score = 4 would be likely to lead to a tangible and viable problem solution	score = 3 have a potential to lead to a tangible and viable problem solution	score = 2 have some potential to lead to a tangible and viable problem solution	score = 1 have little potential to lead to a tangible and viable problem solution	score = 0 cannot lead to a tangible and viable problem solution
Key Vocabulary • "tangible and viable" means it can be touched (made in real life) and capable of working successfully (meets your design requirements)						
Notes • The design requirements should be very clear and comprehensive so that they springboard you toward a solution. • You are NOT looking to describe or really even formulate the specific design at this stage. That is for Elements D-F. • Take care to not list so many requirements that it becomes unlikely to create a viable solution—a viable solution must meet ALL requirements, so only list requirements that your successful design will meet. • You may revisit this element as needed to reflect your design evolution.						
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 no primary stakeholder groups.
Notes • Describe in detail how these design requirements will meet the needs of the clearly identified stakeholder groups. • Document how your stakeholders and stakeholder groups have seen and agree with your design requirements. • Reminder: Primary stakeholder groups need to be identified clearly in Element A						

Inexpensive is a medium priority on the account of the materials (wood, machinery, CADD softwares) being accessible for usage. This application of the inexpensive requirement is a group goal—to not surpass the budget quota and to combat the inaccessibility factor of assistive technology. Many parents, hospitals, and under-funded school districts struggle in a collective battle against a saturated market of expensive toys/devices (Langtree). In the context of providing development-specific technology to a group of children facing disabilities, we want our device to be cost effective. This goal ensures that, in real world application, our device would be available to any buyer.

Marble Track Wall – Design Requirements

1. Purpose and Goals

- The marble track wall should provide interactive entertainment that encourages sensory engagement, fine motor skills, and cause-and-effect learning.

- It should be safe, calming, and inclusive for students with a range of physical and cognitive abilities.
- Be durable enough so kids can really beat on it and it still functions.

2. Safety and Accessibility

- No small removable parts that could be a choking hazard (use marbles larger than 1.75 inches in diameter).
- Smooth, rounded edges on all materials to prevent injuries.
- Constructed from non-toxic, durable materials (e.g., finished wood, acrylic, or high-grade plastic).
- Wheelchair-accessible height (bottom section no higher than 24" from the ground).
- All components should be securely attached to prevent detachment or falling.
- Easy to wipe and disinfect, with minimal crevices for dirt buildup.

3. Dimensions and Structure

- Wall area: approximately 3 feet wide and 3 feet tall.
- Bolted straight to the wall or floor so kids will not be able to move it.

4. Functionality and Interaction

- Tracks should allow marbles to roll smoothly from top to bottom without obstruction.
- Include interchangeable track pieces (ramps, funnels, spinners, chutes, bridges).
- Incorporate sensory features:
 - Visual: bright colors, transparent tubes, and motion.
 - Auditory: gentle clicking, rolling, or chime sounds.
 - Tactile: different textures on surfaces and handles.
- Magnetic or peg-based attachment system for easy rearrangement.

5. Educational and Therapeutic Features

- Supports fine motor development (attaching tracks, releasing marbles, moving pieces).
- Optionally include light-up or sound-trigger features when marbles pass certain points.
- Allows for group play to encourage social interaction and teamwork.

6. Maintenance and Durability

- Must withstand daily classroom use and frequent touching.
- Components should be easy to replace if damaged.
- Use washable, water-resistant materials.
- A manual/guide for instructors if anything were to break

7. Aesthetics

- Bright but non-overstimulating colors (e.g., pastel tones or nature-inspired palette).
- Visually appealing design that complements classroom decor.

Dont grade

Qualities of Design with Metrics/Targets/Tolerance:

General QODs	1	2	3	4
Accessible	Zero components of the assistive technology aid in motor-skill development	A few components that are okay demonstrations of motor-skills.		Kids of all disabilities and impairments can play with the technology with
Aesthetics	The device is bare: no wood finish, no paint, no design. There is no pop of color that helps with pattern recognition 	There is color to the assistive technology but there is no meaning. The color doesn't aid in motor-skills development. AKA, the color is just for show 	A noticeable color gradient that aids in skill development but there are little embellishments to the structure of the device as a whole. 	There is purpose behind the color choice, design, layout, size, shape, material. Designed with the intention of motor-skill enhancement. 
Durable	The device's structure breaks within one use . Examples of breakage could include broken wiring components, wall/trim/corner/edge are damaged, any type of plexi or acrylic is			Besides the possibility of a mechanical portion, the structural integrity remains intact for 5+ years

	chipped.			
Engaging	A child can not be entertained for at least a minute . The child either walks away, gets side tracked because of lack of engagement, or gets frustrated from the toy. ✓	Takes 5 minutes to become disengaged from the device. ✓	In a period between 5 to 15 minutes , the child becomes uninterested in the device. ✓	The kid(s) remains engaged by the toy for the entire duration of the intended “play time” ✓
Inexpensive	The budget of ____ is passed by an additional + \$25. ✓	The additional money spent after passing the budget quota falls in the range of \$24-\$10. ✓	The price sum of the materials passed the budget by \$9-\$1. ✓	Spent the exact budget or in a better scenario, spent below the budget. ✓
Safe	Visible wiring, rough surface, sharp edges and corners, edible pieces. Also can't be disinfected ✓	Disinfection is possible but is required after each use. No sharp corners or edges but the device is still throwable		No reported injuries caused by the device. All electrical components are out of touch. In essence, the teacher stamp of approval. ✓
Sensory	No sound, no lighting, no color, no embellishments meant to serve a greater purpose for those with	1 to 3 embellishments, range in anything from visual, smell, sound, and touch. ✓	4-5 embellishments to help children with motor-skill development. ✓	6+ sensory mechanisms that provide sound, tactile use, visual aid, etc... ✓

	disabilities. 			
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